

ANSWER KEY

EET 203

Electrical and Electronics Measurement

1. Measurement is the act, or the result, of a quantitative comparison between a given quantity and a quantity of the same kind chosen as a unit. The result of the measurement is expressed by a predefined scale or a number representing the ratio between the unknown quantity and the standard.

The operation requires measurement of proper performance. Provides the quantitative information on the actual state of physical variables & processes.

2. Drift is the gradual shift in the indication or record of the instrument over an extended period of time, during which the true value of the variable does not change. No drift means that with a given input the measured values do not vary with time. Drift is an undesirable quality in instruments.

Zero drift or bias describes the effect where the zero reading of an instrument is modified by a change in ambient conditions. This causes a constant error that exists over the full range of measurement of the instrument. Sensitivity drift (also known as scale factor drift) defines the amount by which an instrument's sensitivity of measurement varies as ambient conditions change. It is quantified by sensitivity drift coefficients that define how much drift there is for a unit change in each environmental parameter that the instrument characteristics are sensitive to. Many components within an instrument are affected by environmental fluctuations, such as temperature changes: for instance, the modulus of elasticity of a spring is temperature dependent.

3.

Let, i_f = current in the fixed coil

i_m = current in the moving coil

i = load current

v = load voltage

T_{in} = instantaneous value of the deflecting torque

p = instantaneous power

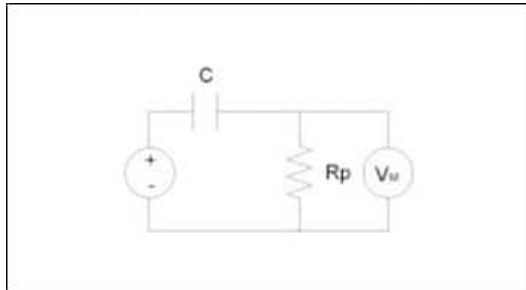
$$\begin{aligned} T_{in} &\propto i_f i_m \\ \text{But since } i_f &\propto i \text{ and } i_m \propto v \\ T_{in} &\propto vi \propto p \end{aligned}$$

4.

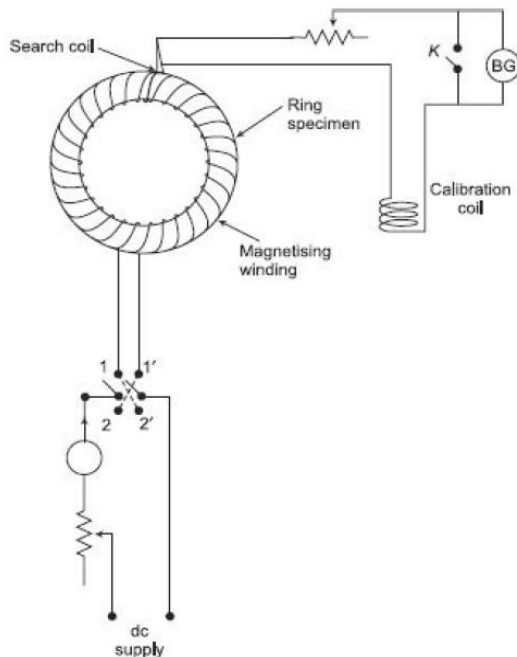
Adjustable copper shading rings are provided on the central limb of the shunt magnet to make

the phase angle displacement between magnetic field set up by shunt magnet and supply voltage is approximately 90 degree. Shading bands made of copper are provided on the central limb of the shunt magnet. are used to bring the flux produced by a shunt magnet exactly in quadrature with the applied voltage.

5. Dielectric of a capacitor is never ideal or perfect. So there will be a small current between terminals of capacitor, either through the dielectric itself, or through other insulating materials. This small current is the leakage current. This is often represented by a resistance parallel with the capacitor, when drawing its equivalent circuit. It is the resistance of the dielectric material itself. IR is measured by leakage current. Knowing the leakage current and applied voltage, the insulation resistance can be calculated based on the ohm's law. There are two basic ways to measure the leakage current. First, apply an ammeter in series with the capacitor and voltage source.



6. Wien's Bridge
7. Flux density can be measured using Ballistic galvanometer connected to search coil.



8. A photodiode is a semiconductor p-n junction device that converts light into an electrical current. The current is generated when photons are absorbed in the photodiode. Photodiodes may contain optical filters, built-in lenses, and may have large or small surface areas. Photodiodes usually have a slower response time as their surface area increases. The common, traditional solar cell used to generate electric solar power is a large area photodiode.

Photodiodes are similar to regular semiconductor diodes except that they may be either exposed (to detect vacuum UV or X-rays) or packaged with a window or optical fiber connection to allow light to reach the sensitive part of the device. Many diodes designed for use specially as a photodiode use a PIN junction rather than a p–n junction, to increase the speed of response. A photodiode is designed to operate in reverse bias.

9. Lissajous figure, also called BOWDITCH CURVE, pattern produced by the intersection of two sinusoidal curves the axes of which are at right angles to each other. If the frequency and phase angle of the two curves are identical, the resultant is a straight line lying at 45° (and 225°) to the coordinate axes. If one of the curves is 180° out of phase with respect to the other, another straight line is produced lying 90° away from the line produced where the curves are in phase (*i.e.*, at 135° and 315°).

Otherwise, with identical amplitude and frequency but a varying phase relation, ellipses are formed with varying angular positions, except that a phase difference of 90° (or 270°) produces a circle around the origin. If the curves are out of phase and differing in frequency, intricate meshing figures are formed.

10. **Advantages of LVDT:**

- LVDT has a low power consumption
- It has a higher sensitive
- It has a ruggedness
- Easy to align and maintain
- It has a wide range
- It has a lower hysteresis loss
- It has a higher measurement range
- It is a frictionless device
- It offers a high resolution which is greater than 10 nm
- It is less in weight
- It is smaller in size
- It is solid and robust, simple in construction
- It has excellent repeatability
- It has a lower output impedance
- It is not affected due to the external environment
- The sliding core does not touch inside the tube and hence it can move without friction

Disadvantages of LVDT:

- Sensitive to a stray magnetic field
- It has large primary voltage produce distortion in an output
- Temperature affects the performance
- It has limited dynamic response
- Vibration due to displacement can affect the performance of the LVDT device
- DC output external demodulator is required
- Large displacement is needed for small output

11. A) The act of this torque/force is opposite to the deflecting torque/force.

When the deflecting and controlling torques are equal in magnitude then the movement will be in definite position or in equilibrium.

Spiral springs or gravity is usually given to produce the controlling torque.

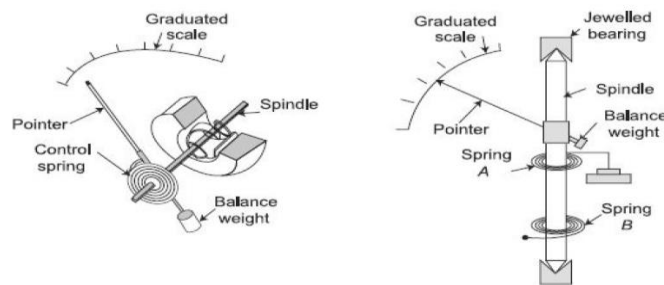
The system which produces the controlling torque is called the *controlling system*.

The functions of the controlling system are

- To produce a torque equal and opposite to the deflecting torque at the final steady position of the pointer in order to make the deflection of the pointer definite for a particular magnitude of current
- To bring the moving system back to its zero position when the force causing the instrument moving system to deflect is removed

► **Spring Control**

A hair-spring, usually of phosphor-bronze attached to the moving system, is used in indicating instruments for control purpose, the schematic arrangement being shown in fig



Thus, the deflection is proportional to the current throughout the scale.

So linear scale

► **Gravity Control**

In a gravity-controlled instrument, a small weight is attached to the moving system in such a way that it produces a restoring or controlling torque when the system is deflected.

- The controlling torque, when the deflection is ϑ , is $Wl \sin \vartheta$, where W is the control weight and l its distance from the axis of rotation of the moving system, and it is, therefore, proportional only to the *sine* of the angle of deflection .

Non linear scale.

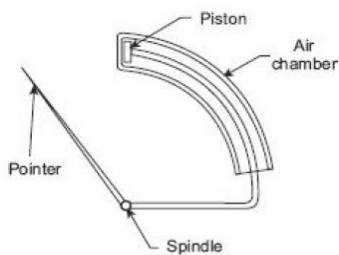
B) When the deflecting torque is much greater than the controlling torque, the system is called underdamped. If the deflecting torque is equal to the controlling torque, it is called *critically damped*. When deflecting torque is much less than the controlling torque, the system is under overdamped condition.

There are three systems of damping generally used. These are as follows:

► **Air-friction damping**

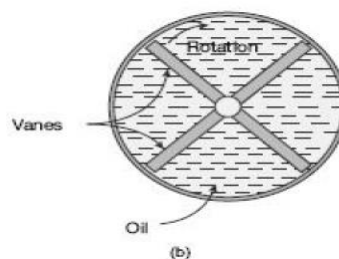
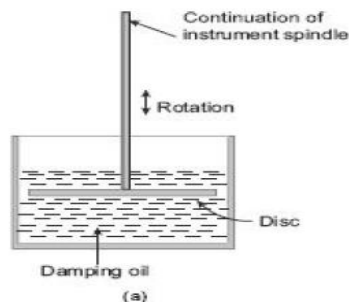
- ▶ Fluid-friction damping
- ▶ Eddy-current damping
- ▶ Air Friction damping

In this method, a light aluminum piston is attached to the moving system and moves in an air chamber closed at one end, as shown in Figure. The cross-section of this chamber may be either circular or rectangular. The clearance between the piston and the sides of the chamber should be small and uniform. If the piston is moving rapidly into the chamber, the air in the closed space is compressed and the pressure opposes the motion of the piston (and, therefore, of the whole moving system). If the piston is moving out of the chamber rapidly, the pressure in the closed space falls, and the pressure on the open side of the piston is greater than that on the opposite side. Motion is thus again opposed.



- ▶ Fluid – Friction

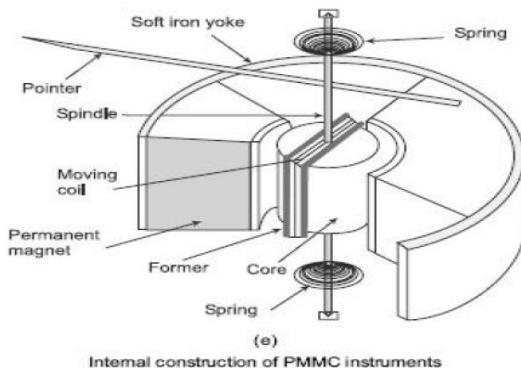
In this type of damping, a light vane, attached to the spindle of the moving system, dips into a pot of damping oil and should be completely submerged by the oil. This is illustrated in Figure (a). The frictional drag in the disc is always in the direction opposing motion. There is no friction force when the disc is stationary. In the second system [Figure (b)], increased damping is obtained by the use of vanes.



- ▶ Eddy Current Damp;ing

When a sheet of conducting material moves in a magnetic field so as to cut through lines of force, eddy currents are set up in it and a force exists between these currents and the magnetic field, which is always in the direction opposing the motion. The force is proportional to the magnitude of the current and to the strength of the field. The magnitude of the current is proportional to the velocity of movement of the conductor, and thus, if the magnetic field is constant, the damping force is proportional to the velocity of the moving system and is zero when there is no movement of the system.

12. A)



The principle on which a Permanent Magnet Moving Coil (PMMC) instrument operates is that a torque is exerted on a current-carrying coil placed in the field of a permanent magnet. A PMMC instrument is shown in Figure. The coil C has a number of turns of thin insulated wires wound on a rectangular aluminum former F. The frame is carried on a spindle S mounted in jewel bearings J1, J2. A pointer PR is attached to the spindle so that it moves over a calibrated scale. The whole of the moving system is made as light in weight as possible to keep the friction at the bearing to a minimum.

The coil is free to rotate in air gaps formed between the shaped soft-iron pole piece (pp) of a permanent magnet PM and a fixed soft-iron cylindrical core IC. The core serves two purposes; (a) it intensifies the magnetic field by reducing the length of the air gap, and (b) it makes the field radial and uniform in the air gap. Thus, the coil always moves at right angles to the magnetic field. A soft-iron yoke (Y) is used to complete the flux path and to provide shielding from stray external fields.

B) The force produced in the coil is given by

$$F = BIL \sin \Theta$$

When $\Theta = 90^\circ$

For N turns, $F = NBIL$

Torque produced $T_d = F \times \text{perpendicular distance}$

$$T_d = NBIL \times b = BINA$$

$$T_d \propto I$$

TC is the control torque and k_s is the spring constant. At final steady state position, Control torque = Deflecting torque

$$\therefore T_c = T_d$$

$$k_s \theta = k_1 i$$

$$\text{or } \boxed{\theta = \frac{k_1}{k_s} i = ki}$$

$$\text{where } k = \frac{k_1}{k_s} = \text{constant}$$

Therefore angle of deflection is proportional to the deflection torque

13. Ratio Error

Transformation ratio is not equal to CT turn ratio (I_2 / I_1 not equal to N_1 / N_2) because of magnetizing and core loss components of the exciting current. It also gets affected due to the secondary current and its power factor.

$$\% \text{ Ratio error} = \frac{\text{nominal ratio} - \text{actual ratio}}{\text{actual ratio}} \times 100$$

$$\% \text{ Ratio error} = \frac{K_n - R}{R} \times 100$$

Phase Angle Error

Ideally the angle between the primary current and secondary current should be 180 degree. But there is some deviation from 180 degree. This deviation is called phase angle error. As can be seen from phasor, this departure is by an angle θ . Phase difference between primary current and secondary current = θ

14.

A) Errors In Induction-type Energy Meters and Their Compensation:

Phase-angle Error

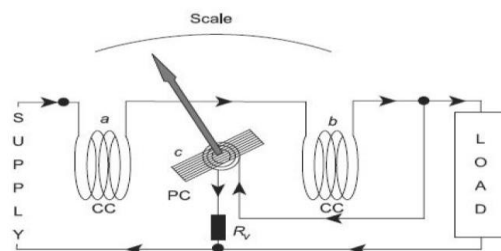
Error due to Friction at Light Loads

Error due to Change in Temperature

Error due to overload

Error due to voltage variation

B)



Let, i_f = current in the fixed coil

i_m = current in the moving coil

i = load current

v = load voltage

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p = instantaneous power

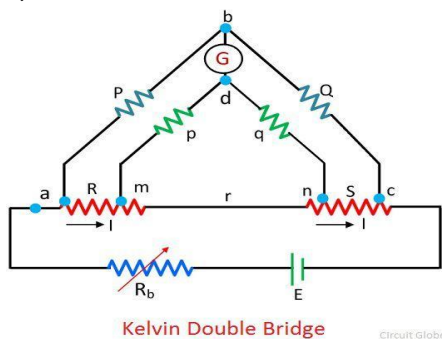
$$T_{in} \propto i_f i_m$$

But since $i_f \propto i$ and $i_m \propto v$

$$T_{in} \propto vi \propto p$$

15.

A)



$$R = \frac{P}{Q} \cdot S + \frac{pr}{p+q+r} \left[\frac{P}{Q} - \frac{p}{q} \right]$$

$$R = \frac{P}{Q} \cdot S$$

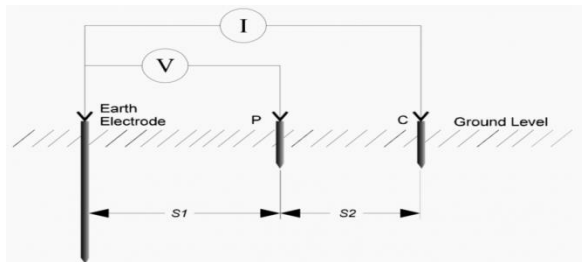
B) On these factors the resistance of earthing system are depends-

Shape and material of electrode or electrodes used. Depth in the soil at which the electrodes are buried.

- Specific resistance of soil surrounding and in the neighborhood of electrodes. The specific resistance of the soil varies from one type of soil to another type of soil. Means its not constant. The specific resistance of soil varies between wide limits and very much depended open its moisture contain.

The 3-point method, called the **“fall of potential” method**, comprises the Earth Electrode to be measured and two other electrically independent test electrodes, usually labelled P (Potential) and C (Current).

- These test electrodes can be of lesser “quality” (higher earth resistance) but must be electrically independent of the electrode to be measured.



16.

A)

The Schering bridge use for measuring the low permeability of the dielectric material. The relative permeability shows the ability of the material for the formation of the magnetic field. It is calculated with the help of the capacitance and dimension of the electrodes. The relative permeability of the parallel plate arrangement is expressed as

$$\epsilon_r = \frac{C_s d}{\epsilon_0 A}$$

C_s – the measured value of capacitance by considering the specimen as a dielectric.

d – Spacing between electrodes

A – Effective area of electrodes.

ϵ_0 – permittivity of free space

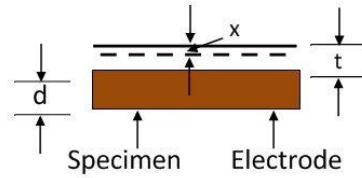
The other method for calculating the relative permittivity of the specimen is explained below. The relative permittivity of the specimen depends on the thickness of the specimen and the spacing between them and the electrode.

Let, C – capacitance between the electrode and specimen

A – area of electrodes

d – the thickness of the specimen

t – the gap between specimen and electrode.



Measurement of Relative Permeability

Circuit Globe

x – reduction in separation between the specimen and electrode.

C_s – capacitance of specimen.

C_0 – Capacitance between the space because of specimen and electrode.

C – effective capacitance of C_s and C_0 .

The capacitance between the specimen and the electrode is expressed as

$$C = \frac{C_s C_0}{C_s + C_0} = \frac{\left(\frac{\epsilon_r \epsilon_0 A}{d}\right) \cdot \left(\frac{\epsilon_0 A}{t}\right)}{\left(\frac{\epsilon_r \epsilon_0 A}{d}\right) + \left(\frac{\epsilon_0 A}{t}\right)} = \frac{\epsilon_r \epsilon_0 A}{\epsilon_r t + d}$$

When we reduce the specimen and the spacing is again adjusted for obtaining the same value of capacitance, the expression for reducing specimen is.

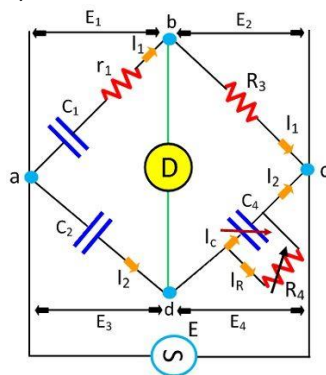
$$C = \frac{\epsilon_r \epsilon_0 A}{\epsilon_r t + d}$$

$$\frac{\epsilon_0 A v}{1 + d - x} = \frac{\epsilon_r \epsilon_0 A}{\epsilon_r t + d}$$

$$\epsilon_r = \frac{d}{d - x}$$

The insulating properties of electrical cables and equipment can also be measured through the Schering bridge.

B)



Low Voltage Schering Bridge

Circuit Globe

When the bridge is in the balanced condition, zero current passes through the detector, which shows that the potential across the detector is zero. At balance condition

$$Z_1/Z_2 = Z_3/Z_4$$

$$Z_1 Z_4 = Z_2 Z_3$$

$$\left(r_1 + \frac{1}{j\omega C_1}\right) \left(\frac{R_4}{1 + j\omega C_4 R_4}\right) = \frac{1}{j\omega C_2} \cdot R_3$$

$$\left(r_1 + \frac{1}{j\omega C_1}\right) R_4 = \frac{R_3}{j\omega C_2} (1 + j\omega C_4 R_4)$$

$$r_1 R_4 - \frac{j R_4}{\omega C_1} = -j \frac{R_3}{\omega C_2} + \frac{R_3 R_4 C_4}{C_2}$$

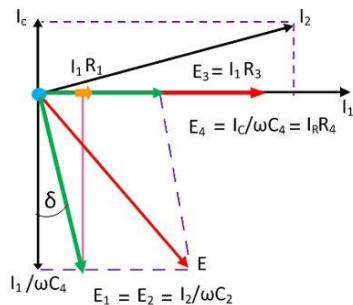
Equating the real and imaginary equations, we get

$$r_1 = \frac{R_3 C_4}{C_2} \dots \dots \dots equ(1)$$

$$C_1 = C_2 \left(\frac{R_4}{R_3}\right) \dots \dots \dots equ(2)$$

The equation (1) and (2) are the balanced equation, and it is free from the frequency.

The dissipation factor obtains with the help of the phasor diagram. The dissipation factor determines the rate of loss of energy that occurs because of the oscillations of the electrical and mechanical instrument.



Phasor Diagram of Low Voltage Schering Bridge
Circuit Globe

$$D_1 = \tan \delta = \omega C_1 r_1 = \omega (C_1 r_1) = \omega (C_2 R_4 / R_3) \times (R_3 C_4 / C_2)$$

$$D_1 = \omega C_4 R_4$$

By the help of the above equation, we can calculate the value of $\tan \delta$ which is the dissipation factor of the Schering bridge.

17. **After demagnetizing the test is started by setting the magnetizing current to its lowest test level. At that time, the galvanometer switch is closed, the iron specimen is brought into a reproducible cyclic magnetic state by throwing the reversing switch S backward and forward about twenty times. The K switch is now opened and the value of flux corresponding of this value of H is measured by reversing the switch S and recording the throw of galvanometer. The value of flux density corresponding to this H can be calculated by dividing the flux by the cross-section area of the specimen. This procedure is repeated for various values of H to maximum testing point. The B-H curve may be plotted from the measurement values of B corresponding to the various values of H.**
Step-by-step Method

In this test the magnetization winding of the ring specimen is supplied from a potential divider having few tapping's. The tapping's are arranged so that the magnetization force H may be increased in suitable number of steps, up to desired maximum value. Before testing the specimen, it must be demagnetised. Keeping the switch S2 on the tap 1 position, the switch S1 is closed. Current flow in the magnetising winding

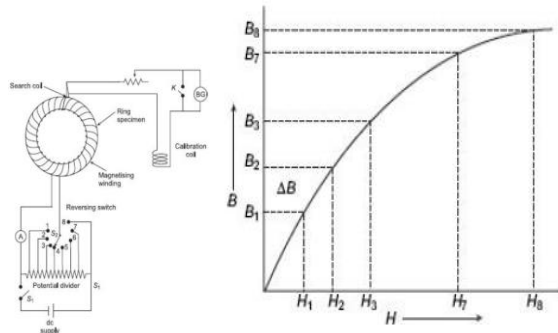
sets up a magnetizing force, H_1 which in turns increases the flux density in the specimen, from zero to some value B_1 , and corresponding throw of galvanometer is observed.

The B_1 value can be calculated from the throw of the galvanometer.

H_1 value may be calculated from the reading of the ammeter, connected to the magnetizing winding circuit.

The magnetizing force is then increased to some value H_2 by switching S_2 suddenly to the tapping 2 and the corresponding increase in flux density ΔB is determined from the throw of the galvanometer. Then the flux density B_2 corresponding to magnetizing force H_2 is calculated as $B_2 = B_1 + \Delta B$.

The process is repeated for other values of H up to the maximum point and the complete B - H curve is thus drawn



B) A thermistor (or thermal resistor) is defined as a type of resistor whose electrical resistance varies with changes in temperature. Although all resistors' resistance will fluctuate slightly with temperature, a thermistor is particularly sensitive to temperature changes.

.They are compact, rugged and inexpensive

They have good stability when properly aged

The response time of thermistors can vary from a fraction of a second to minutes, depending on the size of the detecting mass and thermal capacity of the thermistor

18. A) Temperature sensors detect a change in a physical parameter such as resistance or output voltage that corresponds to a temperature change. There are two basic types of temperature sensing:

Contact temperature sensing requires the sensor to be in direct physical contact with the media or object being sensed. It can be used to monitor the temperature of solids, liquids or gases over an extremely wide temperature range.

Non-contact measurement interprets the radiant energy of a heat source in the form of energy emitted in the infrared portion of the electromagnetic spectrum. This method can be used to monitor non-reflective solids and liquids but is not effective with gases due to their natural transparency.

The Thermostat

The Thermostat is a contact type electro-mechanical temperature sensor or switch, that basically consists of two different metals such as nickel, copper, tungsten or aluminium etc, that are bonded together to form a Bi-metallic strip. The different linear expansion rates of the two dissimilar metals produces a mechanical bending movement when the strip is subjected to heat.

The bi-metallic strip can be used itself as an electrical switch or as a mechanical way of operating an electrical switch in thermostatic controls and are used extensively to control hot

water heating elements in boilers, furnaces, hot water storage tanks as well as in vehicle radiator cooling systems.

The Thermistor

The Thermistor is another type of temperature sensor, whose name is a combination of the words THERM-ally sensitive res-ISTOR. A thermistor is a special type of resistor which changes its physical resistance when exposed to changes in temperature. Thermistors are generally made from ceramic materials such as oxides of nickel, manganese or cobalt coated in glass which makes them easily damaged. Their main advantage over snap-action types is their speed of response to any changes in temperature, accuracy and repeatability.

Most types of thermistor's have a *Negative Temperature Coefficient* of resistance or (*NTC*), that is their resistance value goes DOWN with an increase in the temperature, and of course there are some which have a *Positive Temperature Coefficient*, (*PTC*), in that their resistance value goes UP with an increase in temperature.

B)

Definition: The Photovoltaic cell is the semiconductor device that converts the light into electrical energy. The voltage induced by the PV cell depends on the intensity of light incident on it. The name Photovoltaic is because of their voltage producing capability.

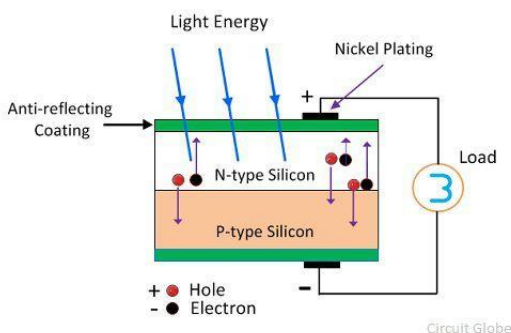
The electrons of the semiconductor material are joined together by the covalent bond. The electromagnetic radiations are made of small energy particles called photons. When the photons are incident on the semiconductor material, then the electrons become energised and starts emitting.

The energised electron is known as the Photoelectrons. And the phenomenon of emission of electrons is known as the photoelectric effect. The working of the Photovoltaic cell depends on the photoelectric effect.

Construction of Photovoltaic Cell

The semiconductor materials like arsenide, indium, cadmium, silicon, selenium and gallium are used for making the PV cells. Mostly silicon and selenium are used for making the cell.

Consider the figure below shows the constructions of the silicon photovoltaic cell. The upper surface of the cell is made of the thin layer of the p-type material so that the light can easily enter into the material. The metal rings are placed around p-type and n-type material which acts as their positive and negative output terminals respectively.



The multi-crystalline or monocrystalline semiconductor material make the single unit of the PV cell. The mono-crystal cell is cut from the volume of the semiconductor material. The multicell are obtained from the material which has many sides.

The output voltage and current obtained from the single unit of the cell is very less. **The magnitude of the output voltage is 0.6v, and that of the current is 0.8v.** The different combinations of cells are used for increasing the output efficiency. There are three possible ways of combining the PV cells.

19.

A)

The most widely used inductive transducer to translate the linear motion into electrical signals is the linear variable differential transformer (LVDT).

The transformer consists of a single primary winding 'p' and two secondary windings s1 and s2 wound on a cylindrical former.

A sinusoidal voltage of amplitude 3 to 15 volt and frequency 50 to 20000 hz is used to excite the primary.

- The two secondaries have equal number of turns and are identically placed on either side of the primary winding.

The primary winding is connected to an alternating current source.

A movable soft-iron core is placed inside the former. The displacement to be measured is applied to the arm attached to the soft iron core.

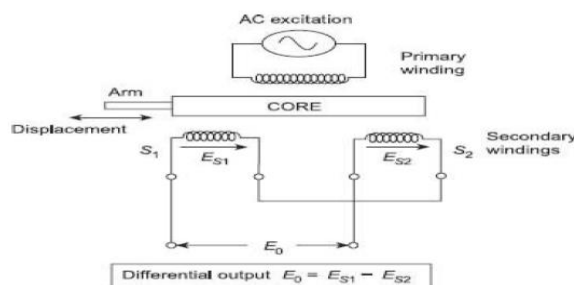
In practice the core is made of high permeability, nickel iron. This is slotted longitudinally to reduce eddy current losses.

The assembly is placed in a stainless steel housing to provide electrostatic and electromagnetic shielding. The frequency of ac signal applied to primary winding may be between 50 hz to 20 khz.

Since the primary winding is excited by an alternating current source, it produces an alternating magnetic field which in turn induces alternating voltages in the two secondary windings.

The output voltage of secondary s1 is e_{s1} and that of secondary s2 is e_{s2} . In order to convert the outputs from S1 and S2 into a single voltage signal, the two secondary S1 and S2 are connected in series opposition as shown in figure.

Differential output voltage $e_0 = e_{s1} - e_{s2}$



Advantages

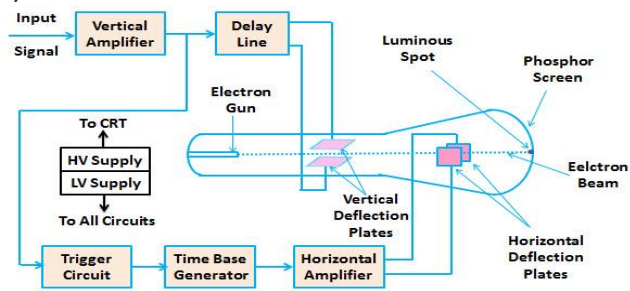
Linearity is good up to 5 mm of displacement.

- Output is rather high. Therefore, immediate amplification is not necessary.
- Output voltage is stepless and hence the resolution is very good.
- Sensitivity is high (about 40 v/mm).
- It does not load the measurand mechanically.
- It consumes low power and low hysteresis loss also.

b)

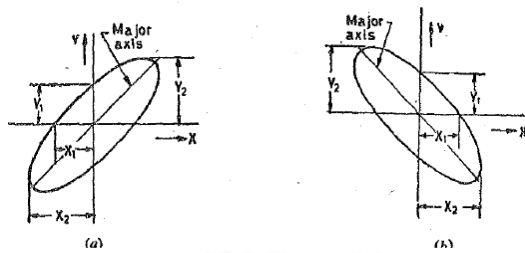
20.

A)



Block Diagram of Cathode Ray Oscilloscope (CRO)

B)



$$\sin \phi = \frac{Y_1}{Y_2}$$

$$\phi = 180^\circ - \sin^{-1} \frac{Y_1}{Y_2}$$